Cloud Task Scheduling: Section 4.2

Replacement for Section 4.2 only, with Figures 6–8. Equation (3) refers to the separately supplied Cost block.

4.2 Experimental Series 2. Cloud Task Scheduling

4.2.1 Experimental Settings and Protocol

The primary cloud comparison uses 50 heterogeneous VMs and $N = 100, 200, \ldots, 1000$ independent tasks. Workloads are uniform integers from $[1000, 20000]$ MI and default processing rates are uniform integers from $[500, 2000]$ MIPS. Seeds 43–72 define 30 paired scenarios per scale, giving 300 primary blocks. Nine additional scales, $N = 2000, 3000, \ldots, 10000$, extend the comparison. Separate task and VM random streams generate each scenario, with VM pools shared across scales within a seed. All search fitness is analytical.

Every available VM is reserved from time zero through the analytical makespan and billed continuously at its synthetic rate. Parameters are $r_{\min} = 0.05$, $r_{\max} = 0.20$ units per second and $\epsilon_\nu = 1$ MIPS. The energy proxy retains $P^{\max} = 250$ W, $\eta_P = 0.6$, $\gamma = 3$, and 3600 seconds per hour. All fixed-reference floors are 10^{-9} in the corresponding units and are inactive in the saved scenarios. Nominal weights are $(0.25, 0.25, 0.25, 0.25)$, giving effective weights $(0.50, 0.25, 0.25)$ under Eq. (3). Local admissibility tolerance is 10^{-12}.

LSCBO, adapted CBO, PSO, GWO, WOA, and adapted AOA use 100 uniformly random population members without a deterministic injected member. The recorded iteration setting is 100 and the nominal candidate allowance is $B = 10000$. Termination follows evaluation accounting, not 100 required full sweeps. The population and primary scale settings are inspired by ERTH [1]; its cost expression, normalization, algorithm, and results are not reproduced. Greedy-LPT supplies the shared fixed normalization references, assigning tasks in descending workload order to the earliest projected finish, with original-index and smallest-VM-index tie breaking. It is not injected into the random populations or displayed as a population competitor. The released main CSV retains 3420 population-method rows and 570 reference-only rows; these reference records support normalization and audit.

The implemented CBO and Lévy parameters are retained. Cloud coordinates span $[0, 1]$, with the 10^{-10} Lévy safeguard. PSO uses $w = 0.9 \to 0.4$, $c_1 = c_2 = 2$, initial velocity $0.1\mathcal{N}(0, 1)$, and clipping to $[-0.2, 0.2]$. GWO and WOA use $a = 2 \to 0$; WOA has spiral coefficient 1 and branch probability 0.5. Adapted AOA uses $\alpha = 5$, $\mu = 0.499$, and MOA from 0.2 to 1. Cloud random streams use `java.util.Random`, derived from scenario, experiment, variant, and profile identifiers.

LSCBO reserves 600 core candidates and 9400 pair trials, with 400 trials per local round. Initialization consumes 100 core evaluations and core proposals consume 500. Other population comparators use all 10000 candidates in their continuous core. One local-entry reevaluation makes 10001 internal LSCBO scoring calls, or 10002 including the common reference evaluation. At most $\lceil 9400/400 \rceil = 24$ local rounds can commit

an exchange, touching at most 48 distinct tasks. The retained archived local sampler and decoder are unchanged by the cost revision.

The new analytical dataset contains 7050 rows: 3990 main rows over 19 scales, 2250 weight rows, and 810 capacity rows. Figures 6 and 7 use only the 2100 primary main rows. Python independently recomputes all four criteria and fitness from every stored assignment and scenario. The primary 2100 rows reproduce the selected-model screening results exactly at stored precision. Main uncertainty uses 30 complete seed clusters, 50000 percentile-bootstrap resamples, and PCG64 seed 20260912. The seed is the resampling unit; scales are retained within each cluster.

Exploratory selection and additional-seed confirmation. Nine cost candidates were screened in two exploratory stages using seeds 43–72, with optimization repeated under each native definition. The continuous fixed-fleet reservation formula was selected by its mean paired cost advantage against its closest population comparator, PSO. The model and comparator were fixed before testing seeds 103–132. The selected model's mean paired reduction against PSO is 25.1840% on the screening seeds and 25.6809% on the new seeds; the latter 95% seed-cluster interval is 24.6229%–26.6663%, with 300/0/0 wins/ties/losses and the lowest mean cost at all ten scales. Confirmation uses 30000 bootstrap resamples with seed 20261011. The main tables reuse screening seeds, so the confirmation supports the selected cost comparison rather than independently validating every main-table claim.

On those same additional instances, the completion-time control is optimized separately. Scoring the new reservation-optimized LSCBO assignments with the old completion-time cost gives values 1.1079% higher on average than that control, with a 95% seed-cluster interval of 0.9969%–1.2258%. The larger reservation-cost separation therefore reflects the changed charging policy and scalar preference, not an improvement of the earlier cost metric.

Table 4 Cold-start configurations and internal scoring calls. The shared reference evaluation is additional.

Method	Core	Pair trials	Internal calls	Population
LSCBO	600	9400	10001	100
Adapted CBO, PSO, GWO, WOA, AOA	10000	0	10000	100

Sensitivity uses the same complete scheduler and core/local split. Normalization adds one shared reference call per instance outside the population allowance.

4.2.2 Main Comparison

LSCBO has the lowest mean F among the six population methods at all ten primary task scales. Its overall mean is 19.953855, compared with 25.704472–31.134720 for the other population methods and 1 for Greedy-LPT. These are scale-mean rankings, not claims of a win in every paired seed. Table 5 reports the six population methods. Resampling complete seed clusters gives a 95% interval of 19.311615–20.616214 for LSCBO's primary mean F.

Table 5 Cold-start mean reference-normalized objective F over 30 paired seeds at each primary task scale (lower is better). Only the six population methods are displayed; reference construction is defined in Section 4.2.1.

Tasks	LSCBO	CBO	PSO	GWO	WOA	Adapted AOA
100	3.4046	4.5759	5.2214	6.3786	5.4342	6.3717
200	6.5952	10.1033	10.7299	13.1799	11.1067	13.2568
300	9.0984	14.6207	15.0726	17.9760	14.7277	18.0556
400	13.8638	21.1788	21.0465	25.0603	21.1661	25.0765
500	18.3725	27.2336	25.4814	31.1117	26.3336	31.2172
600	20.9774	29.1115	27.5281	32.9948	28.2043	33.1799
700	24.9995	33.7967	31.4594	38.0532	33.1611	38.1118
800	30.4452	40.4519	36.3089	44.7519	38.9461	44.6780
900	34.1787	44.7063	41.1325	49.3416	41.0689	48.9646
1000	37.6033	47.9044	43.0639	52.0647	44.6275	52.4352

Table 6 Cold-start LSCBO relative reductions over the shared fixed reference. Negative reductions indicate worse outcomes. Component reductions are ratios of scale-wise means; F reductions are means of paired instance reductions. Parentheses report paired W/T/L counts (30 seeds).

Tasks	Makespan	Energy	Cost	$\mathcal{I}$	F
100	-38.8736% (0/0/30)	-16.7167% (0/0/30)	-38.7936% (0/0/30)	-802.0890% (0/0/30)	-240.4620% (0/0/30)
200	-40.0549% (0/0/30)	-11.9073% (0/0/30)	-40.0203% (0/0/30)	-1968.8160% (0/0/30)	-559.5157% (0/0/30)
300	-42.0570% (0/0/30)	-11.9373% (0/0/30)	-41.9791% (0/0/30)	-2954.5294% (0/0/30)	-809.8373% (0/0/30)
400	-45.9823% (0/0/30)	-13.4877% (0/0/30)	-46.0063% (0/0/30)	-4763.1286% (0/0/30)	-1286.3771% (0/0/30)
500	-49.1954% (0/0/30)	-14.9795% (0/0/30)	-49.1657% (0/0/30)	-6431.1607% (0/0/30)	-1737.2532% (0/0/30)
600	-50.9174% (0/0/30)	-16.3763% (0/0/30)	-50.8768% (0/0/30)	-7663.3794% (0/0/30)	-1997.7350% (0/0/30)
700	-54.3047% (0/0/30)	-17.7835% (0/0/30)	-54.2718% (0/0/30)	-9236.8181% (0/0/30)	-2399.9517% (0/0/30)
800	-59.0601% (0/0/30)	-19.6090% (0/0/30)	-59.1069% (0/0/30)	-11346.9547% (0/0/30)	-2944.5152% (0/0/30)
900	-58.9449% (0/0/30)	-19.7759% (0/0/30)	-58.9044% (0/0/30)	-12802.3247% (0/0/30)	-3317.8728% (0/0/30)
1000	-61.9330% (0/0/30)	-21.4052% (0/0/30)	-61.8518% (0/0/30)	-14239.4761% (0/0/30)	-3660.3349% (0/0/30)

All four LSCBO criteria are higher than Greedy-LPT in every one of the 300 primary instances. Its mean paired reservation-cost increase over the reference is 50.0934%. Reservation cost and makespan have identical paired relative changes

because the fixed fleet-price sum cancels. The negative reductions in Table 6 are therefore regressions, and the incumbent-relative guard does not establish reference-relative non-regression.

Among the population methods, LSCBO has the lowest mean reservation cost at every primary scale. Its mean paired cost reduction against the fixed PSO comparator is 25.1840%, with 298/0/2 wins/ties/losses on the main seeds. Figure 7(c) shows this separation in absolute synthetic units. For a given seed, Cost is exactly a constant multiple of makespan. Since fleet prices vary by seed, aggregated cost means are price-weighted horizon means; an average curve need not be one globally fixed multiple of the makespan curve. The cost panel is a policy-specific financial interpretation of batch duration, not additional independent optimization evidence.

Figure 6 shows mean normalized objectives on a logarithmic axis. Figure 7 shows absolute component means. The curves and comparison tables display the population methods. Shared normalization references are specified in Section 4.2.1 and retained in the released raw records. There is no cost inset, smoothing, self-relative reference line, or vertical offset.

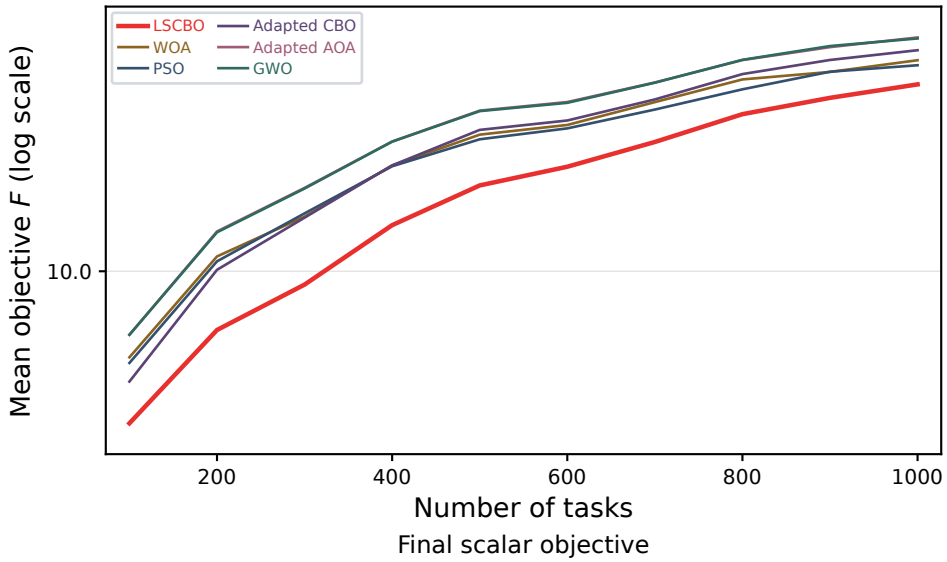


Fig. 6 Cold-start comparison at 100–1000 tasks: 100 random population members, a 10000-candidate allowance, 50 VMs, and 30 paired seeds per scale. Mean normalized F for six population methods is shown on a logarithmic axis under batch reservation charging. Shared reference normalization is defined in Section 4.2.1.

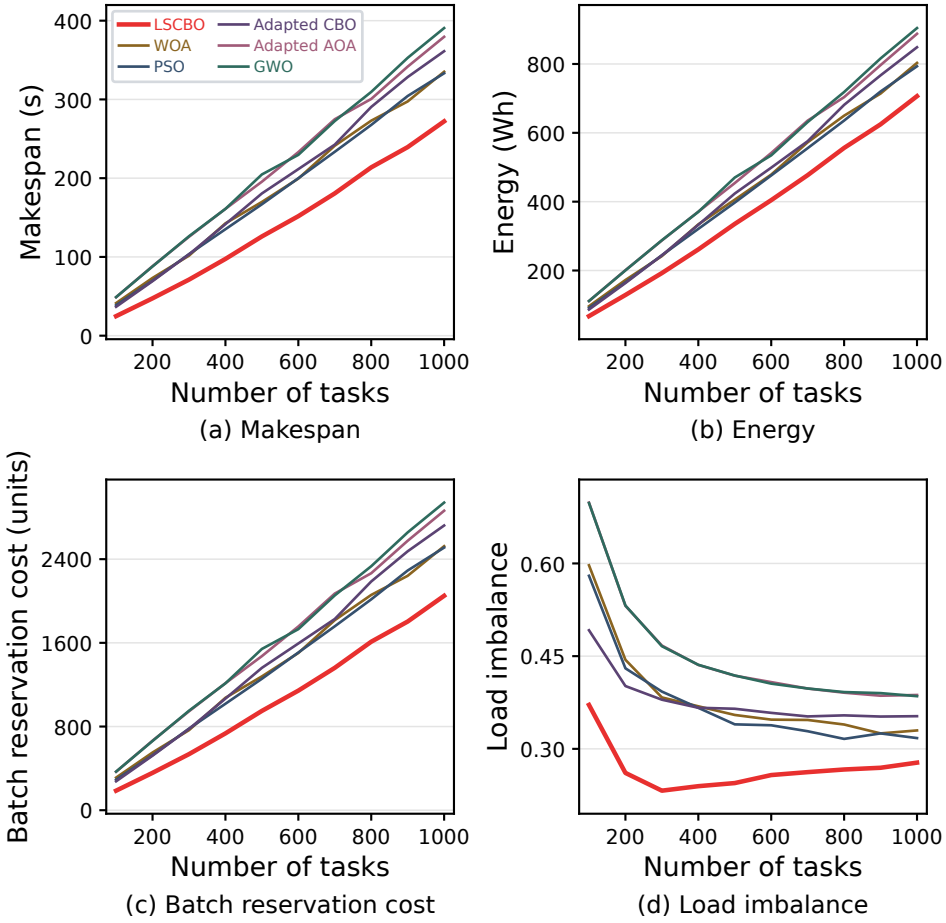


Fig. 7 Cold-start absolute analytical means over 30 paired seeds per scale: (a) makespan, (b) energy, (c) batch reservation cost, and (d) load imbalance. Every VM is charged from zero until common release at batch completion, at synthetic continuous rates. Normalized cost equals normalized makespan within each instance. Each method has the same color, solid line, and width in all panels and the legend. Shared reference normalization is defined in Section 4.2.1; the underlying reference records accompany the raw data.

Table 7 Cold-start absolute analytical outcomes for LSCBO at all ten primary task scales. Each cell gives the mean on the first line and sample standard deviation in parentheses below, over 30 paired seeds (divisor 29). Batch reservation cost uses synthetic units and charges the entire fixed VM fleet through makespan; $\mathcal{I}$ and F are dimensionless.

Tasks	$H_{\max}$ (s)	E (Wh)	C (units)	$\mathcal{I}$	F
	24.803	67.395	186.775	0.37109	3.405
100	(2.915)	(6.494)	(19.354)	(0.05427)	(0.755)
	47.435	128.852	357.246	0.26104	6.595
200	(4.038)	(9.501)	(24.197)	(0.04185)	(1.821)
	71.165	192.498	535.959	0.23232	9.098
300	(5.656)	(13.139)	(31.814)	(0.02779)	(2.146)
	97.459	261.210	734.494	0.23959	13.864
400	(5.659)	(14.676)	(33.540)	(0.02619)	(3.488)
	125.993	335.354	949.328	0.24462	18.373
500	(9.320)	(19.682)	(57.574)	(0.02175)	(5.753)
	151.732	404.373	1143.133	0.25760	20.977
600	(11.150)	(24.897)	(66.586)	(0.02685)	(3.963)
	180.703	476.994	1361.220	0.26233	25.000
700	(12.185)	(28.358)	(64.676)	(0.02528)	(3.893)
	213.712	556.142	1610.678	0.26659	30.445
800	(15.652)	(34.786)	(104.967)	(0.02458)	(5.187)
	239.444	624.647	1803.612	0.26939	34.179
900	(17.131)	(39.423)	(95.282)	(0.02386)	(6.169)
	272.157	706.486	2049.752	0.27769	37.603
1000	(22.448)	(46.948)	(132.202)	(0.02487)	(5.824)

The nine external scales retain the same cost policy and candidate allowance. WOA overtakes LSCBO at $N \in \{4000, 5000, 6000, 7000, 8000, 9000, 10000\}$, lowering scale-wise mean F by 0.50–8.34% relative to LSCBO. LSCBO retains a lower or equal mean than WOA at $N \in \{2000, 3000\}$. The released CSV contains all 19 scales. The large-scale crossover limits the primary ranking.

4.2.3 Direct Greedy-LPT Refinement

The direct-refinement diagnostic uses the same 300 primary scenarios, equal nominal weights, 50 VMs, and batch reservation charging. A separately compiled instrumented diagnostic reproduces LSCBO's five stored outcome metrics in all 300 blocks. The two displayed controls explicitly start from Greedy-LPT and run 9400 or 10000 Pareto-safe pair trials. Their total scoring counts, including reference and local-entry scoring, are 9402 and 10002. This initialization is different from the primary random-start protocol.

The mean objectives are 0.974373 and 0.973684 for the 9400- and 10000-trial controls. Their lower/tied/higher counts against Greedy-LPT are 300/0/0 and 300/0/0. The 10000-trial versus 9400-trial objective counts are 190/110/0. The guard protects the current incumbent's makespan, energy, cost, and imbalance, with the cost constraint mathematically redundant under common release. Changing the final partial round can alter the chosen exchange, so an objective improvement need not dominate every component across separately terminated runs.

Timing uses an Intel Core i7-14650HX Windows 11 machine, 24 logical CPUs, about 32 GB RAM, and Microsoft OpenJDK 21.0.7. One application thread and Serial GC use a 512 MB initial and 2 GB maximum heap. After metric reproduction and two complete warm-up cycles, five repetitions rotate method order. Reported times are means of within-block medians and include reference construction, normalization, and search, but exclude scenario generation, hashing, and CSV output. The analytical main and sensitivity jobs finish before timing begins. Frequency and affinity are not fixed; the measurements are descriptive.

Table 8 Direct Greedy-initialized pair controls under batch reservation charging on 300 paired scenarios. Calls include reference and local-entry scoring; times are means of within-block medians over five warmed repetitions.

Method	Actual calls	Mean F	Mean accepted exchanges	Time (ms)
Greedy + pair search, 9400 trials	9402	0.974373	16.62	32.31
Greedy + pair search, 10000 trials	10002	0.973684	17.34	34.40

4.2.4 Weight Sensitivity

Five nominal profiles are evaluated at $N \in \{100, 300, 500, 800, 1000\}$ with 30 seeds, 50 VMs, and the same 10000-candidate allowance. The profiles are equal weights and makespan-, energy-, cost-, and balance-priority weights; the named priority receives

0.55 and each other criterion 0.15. LSCBO and adapted CBO produce 1500 optimized records; 750 separate reference-only records support normalization, giving 2250 released rows.

The LSCBO means are 19.7924, 12.6171, 12.4294, 12.6148, 41.7855 in that profile order. Each profile's mean exceeds the fixed reference of 1. Absolute F values from different weights express different scalar preferences and should not be ranked as if one objective were unchanged. Table 9 reports every scale separately.

Since normalized cost equals normalized makespan, both nominal makespan and cost priorities reduce to effective weights $(0.70, 0.15, 0.15)$ for makespan, energy, and imbalance. Their stored outcomes differ because the retained implementation derives random streams from profile identifiers, and floating-point evaluation can also differ. These results are not evidence for distinct time and cost preferences. The other effective profiles are $(0.30, 0.55, 0.15)$ for energy and $(0.30, 0.15, 0.55)$ for imbalance. Profile sensitivity remains conditional on these preferences and does not identify a universal optimum or a Pareto front [2].

Table 9 Cold-start sensitivity by task scale: LSCBO objective dispersion and paired reductions over the shared fixed reference.

Priority	Tasks	Mean F ± SD	F reduction (%)	$\mathcal{I}$ reduction (%)	W/T/L
Equal	100	3.436255 ± 0.649934	-243.6255	-876.3666	0/0/30
Equal	300	9.177249 ± 2.097877	-817.7249	-3175.1270	0/0/30
Equal	500	18.790173 ± 5.682190	-1779.0173	-6997.1414	0/0/30
Equal	800	30.108801 ± 4.947311	-2910.8801	-11511.6025	0/0/30
Equal	1000	37.449336 ± 5.367060	-3644.9336	-14436.3790	0/0/30
Makespan	100	2.755092 ± 0.414323	-175.5092	-949.2229	0/0/30
Makespan	300	6.174497 ± 1.303689	-517.4497	-3240.3063	0/0/30
Makespan	500	12.057427 ± 3.482799	-1105.7427	-7118.6393	0/0/30
Makespan	800	18.909402 ± 3.149117	-1790.9402	-11657.6801	0/0/30
Makespan	1000	23.188838 ± 3.173381	-2218.8838	-14482.0037	0/0/30
Energy	100	2.606991 ± 0.375961	-160.6991	-921.5759	0/0/30
Energy	300	6.009445 ± 1.300605	-500.9445	-3213.0211	0/0/30
Energy	500	11.867451 ± 3.599432	-1086.7451	-7083.0945	0/0/30
Energy	800	18.673201 ± 2.953601	-1767.3201	-11603.0906	0/0/30
Energy	1000	22.990036 ± 3.202573	-2199.0036	-14456.3219	0/0/30
Cost	100	2.754723 ± 0.393049	-175.4723	-954.3222	0/0/30
Cost	300	6.175898 ± 1.247661	-517.5898	-3242.0206	0/0/30
Cost	500	12.133824 ± 3.657125	-1113.3824	-7172.9339	0/0/30
Cost	800	18.874548 ± 3.118332	-1787.4548	-11633.2789	0/0/30
Cost	1000	23.135242 ± 3.177319	-2213.5242	-14449.9269	0/0/30
Balance	100	5.777895 ± 1.501541	-477.7895	-842.3375	0/0/30
Balance	300	18.355800 ± 4.318211	-1735.5800	-3129.4018	0/0/30
Balance	500	39.729406 ± 13.032062	-3872.9406	-7009.1862	0/0/30
Balance	800	64.811614 ± 11.168877	-6381.1614	-11565.3267	0/0/30
Balance	1000	80.252641 ± 11.887667	-7925.2641	-14369.5442	0/0/30

Each row uses 30 paired seeds and 50 VMs. Reductions are means of instance-wise percentages; negative values indicate regressions. The shared fixed reference has $F = 1.000000 \pm 0.000000$. W/T/L uses an absolute 10^{-8} tie threshold on F. Equal weights are (0.25,0.25,0.25,0.25); a priority uses weight 0.55 and 0.15 for the other criteria.

4.2.5 VM Heterogeneity

Processing rates are sampled as uniform integers from $[1333, 2000]$, $[400, 2000]$, and $[100, 2000]$ MIPS, labeled approximately 1.5:1, 5:1, and 20:1. Each profile uses $N \in \{500, 1000, 2000\}$, 30 seeds, 50 VMs, and nominal equal weights. The 2000-task cell extends beyond the primary main-comparison range. Each capacity profile also changes the capacity-normalized price mapping; the comparison does not isolate a causal capacity effect.

The LSCBO mean objectives are 24.189983, 50.219427, 53.619678 for the 1.5:1, 5:1, and 20:1 profiles, respectively. All exceed the corresponding Greedy-LPT reference. Table 10 provides the complete scale/profile values. Changing fleet prices rescales absolute Cost, but within a fixed scenario its normalized ratio remains equal to the makespan ratio. Thus a larger absolute bill across profiles is not by itself evidence of a worse independent cost objective.

Table 10 Cold-start sensitivity by task scale: LSCBO objective dispersion and paired reductions over the shared fixed reference.

Capacity ratio	Tasks	Mean $F \pm$ SD	F reduction (%)	$\mathcal{I}$ reduction (%)	W/T/L
1.5:1	500	10.454794 ± 3.966480	-945.4794	-3734.8643	0/0/30
1.5:1	1000	21.415776 ± 4.519804	-2041.5776	-8119.6961	0/0/30
1.5:1	2000	40.699381 ± 6.188546	-3969.9381	-15836.6628	0/0/30
5:1	500	18.582451 ± 4.125177	-1758.2451	-6886.8083	0/0/30
5:1	1000	42.506956 ± 6.672730	-4150.6956	-16409.8914	0/0/30
5:1	2000	89.568875 ± 12.038847	-8856.8875	-35190.3535	0/0/30
20:1	500	22.176768 ± 5.920106	-2117.6768	-8058.3183	0/0/30
20:1	1000	45.268714 ± 11.809057	-4426.8714	-17123.1008	0/0/30
20:1	2000	93.413551 ± 28.442079	-9241.3551	-36171.0771	0/0/30

Each row uses 30 paired seeds and 50 VMs. Reductions are means of instance-wise percentages; negative values indicate regressions. The shared fixed reference has $F = 1.000000 \pm 0.000000$. W/T/L uses an absolute 10^{-8} tie threshold on F. Capacity profiles change both speeds and their capacity-dependent price mapping.

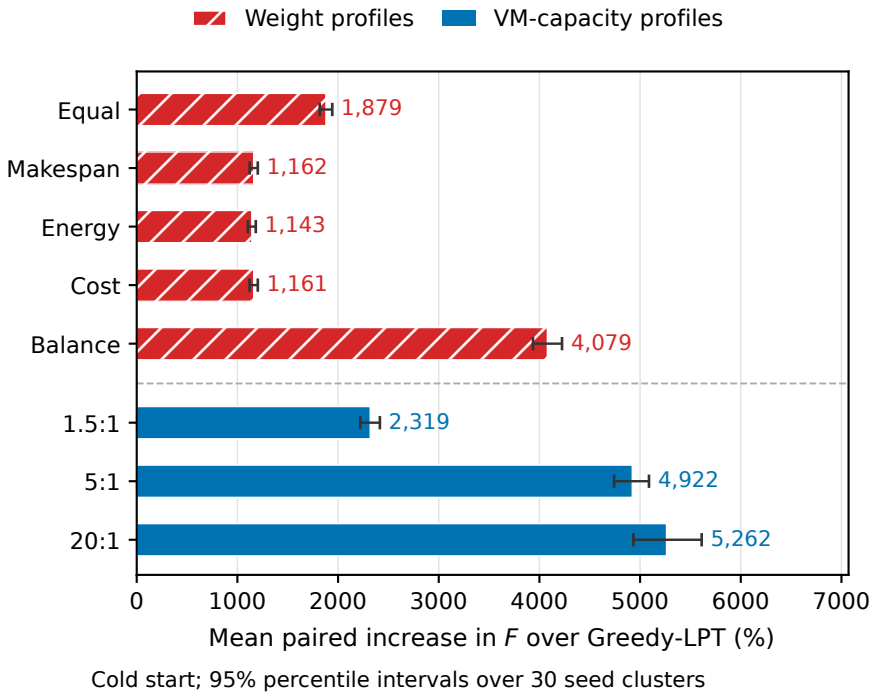


Fig. 8 Cold-start sensitivity under batch reservation charging. Bars show mean paired increases over the shared fixed reference objective, $100(F/F_g - 1)$; lower is better. Five weight profiles retain five scales each, and three VM-capacity profiles retain three scales each. Intervals are 95% seed-cluster percentile intervals from 30 clusters and 50000 resamples (PCG64 seed 20260912). Nominal makespan and cost priorities have identical effective weights but use different profile-derived random streams. The dashed separator distinguishes the two profile families. These are objective increases, not monetary savings.

4.2.6 Descriptive Runtime Analysis

The primary analytical logs give LSCBO scale-wise median runtimes from 19.0–111.0 ms. Table 11 reports LSCBO median and interquartile ranges. These jobs use eight main workers, while a six-worker sensitivity job runs concurrently. They are not controlled speed comparisons. The warmed single-thread direct experiment provides a separate, still platform-specific timing description. Stored quality observations are not multiplied by timing repetitions.

Table 11 Cold-start LSCBO runtime (ms), median [first quartile, third quartile] over 30 seeds. Concurrent logs are descriptive, not controlled timing benchmarks.

Tasks	LSCBO with pair search
100	19.0 [17.00, 21.75]
200	51.0 [23.25, 56.00]
300	63.0 [60.25, 70.00]
400	72.5 [70.00, 79.00]
500	86.0 [81.00, 95.00]
600	94.0 [89.00, 105.50]
700	104.5 [98.00, 111.00]
800	111.0 [105.25, 117.00]
900	79.0 [75.00, 81.75]
1000	86.0 [82.25, 90.75]

References

[1] Qin X, Li S, Tong J, Xie C, Zhang X, Wu F, Xie Q, Ling Y, Lin G (2024) ERTH scheduler: enhanced red-tailed hawk algorithm for multi-cost optimization in cloud task scheduling. Artif Intell Rev 57:328. https://doi.org/10.1007/s10462-024-10945-6

[2] Marler RT, Arora JS (2010) The weighted sum method for multi-objective optimization: new insights. Struct Multidiscipl Optim 41(6):853–862. https://doi.org/10.1007/s00158-009-0460-7